APMM - AI & Automation

Take-home assignment · 3 days

This mirrors the real work: building a small AI agent, calculating recruiter-league winners from usage data, and reading automation code.

Logistics

Time: 3 days from your briefing.

Submit to: product.marketing@foundit.ai

Files you've been given: Sample_Data_for_agent.xlsx (Task 1), **purchase.xlsx** (Task 2), **winner_sea.py** (bonus).

Send back: (1) agent link/screenshots + your system prompt, (2) fRL winners Excel + a short method note, (3) bonus write-up. One email with attachments is fine.

Task 1 · Build a Micro AI Agent on our IT/ITeS talent data

Goal. Build a small agent in Relevance AI (or an n8n workflow) that answers questions from the India IT/ITeS talent-supply data in **Sample_Data_for_agent.xlsx**. This is the same pattern as the live agent our Sales & CS teams use to quote talent availability.

What's in the data: India IT/ITeS candidate-profile counts, broken down several ways:

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INDIA IT/ITeS TALENT SUPPLY  (foundit profile database)
Total profiles: 13.5 Mn  |  12M-active: 6.8 Mn  |  6M-active: 6.0 Mn

By experience : 0-1y 404K · 1-3y 1.1Mn · 3-5y 2.1Mn · 5-10y 4.0Mn ·
            10-15y 3.0Mn · 15y+ 2.9Mn
By city (total): Bengaluru 3.2Mn · Delhi NCR 2.1Mn · Hyderabad 1.9Mn ·
             Pune 1.7Mn · Chennai 1.4Mn · Mumbai 1.3Mn (+ Tier II/III)
By role       : AI/ML Engineer 995K · DevOps 137K · Data Scientist 116K ·
            Cloud Architect 99K · Cybersecurity 76K
Sub-industry  : Information Technology 5.8Mn · Information Services 3.7Mn ·
            IT Management 1.8Mn · Network Security 1.7Mn
```

The full file also splits every cut into 12-month / 6-month active, sourced and registered profiles.

What to do

- Load the data as the agent's knowledge base.
- Write a system prompt: role, scope, answer format, and a guardrail to say "that isn't in my data" when it doesn't know.
- Test 3 questions: (a) *"How many AI/ML Engineer profiles are available in India?"* (answerable) (b) *"Which is larger - Data Scientist or DevOps talent, and by how much?"* (a small comparison) (c) *"What salary should I offer a DevOps engineer in Pune?"* (not in the data → it must decline, not invent a number).

Submit: a shareable link (Preferable) or 2-3 screenshots, your full system prompt, and the 3 sample questions with the agent's answers.

Task 2 · Calculate the fRL Winners

Goal. From the **purchase.xlsx** usage file (Jan 2026, ~17,000 rows), work out the Recruiter-League (fRL) winners by applying every rule below. Use Excel (pivots/filters/formulas) or Python, your choice. Getting the rules right matters more than the tool.

Key columns: company_name, company_type, service_channel, login, email, account_type, PC (search / profile consumption), OC (outbound / email campaign consumption), JP (jobs posted), pc_start_date, pc_end_date.

Fixed settings

- Reference date (active cut-off): 20 February 2026.
- New-account window: 1 October 2025 - 31 January 2026.
- MVP: top 5 per user type.

Metric → award mapping

- Search Smasher = highest PC (search / profile consumption).
- Campaign Captain = highest OC (outbound / email campaign consumption).
- Posting Champion = highest JP (jobs posted).

Rule 1 - Remove ineligible rows first. Drop a row if ANY of these is true:

- account_type is free_trial or test_job.
- login contains scrape, _jobs, or ftp.
- service_channel contains scrape.
- company_name contains immigration, ankit sharma proprietor, or freelancer.
- pc_end_date is before 20 February 2026 (expired account).
- The company already won in a previous cycle (old-winners list — apply only if that list is provided).

Rule 2 - Tag each account New or Existing.

- New = pc_start_date falls within 1 Oct 2025 – 31 Jan 2026.
- Existing = everything else (a blank start date counts as Existing).

Rule 3 - Category winners (do this for each of PC, OC, JP).

- Rank eligible accounts by that metric, highest first, and look at the top 30.
- Pick the #1 New account and the #1 Existing account.
- Result: 1 New + 1 Existing per category → up to 6 category winners.

Rule 4 - MVP (the all-rounder).

- Keep only accounts active in all three: PC > 0 AND OC > 0 AND JP > 0.
- MVP Score = PC + OC + JP.
- Rank separately within New and within Existing; take the top 5 of each → up to 10 MVPs.

Output. One Excel sheet with these columns:

> Source | Category | User Type | MVP Rank | MVP Score | Company Name |
> Company Type | Service Channel | Login | Email | PC | OC | JP |
> Start Date | End Date

Note: *the sample file is the "Purchase" source. In production the same rules also run on a "Bridge" (renewals) source - you only need Purchase here.*

Submit: the winners Excel + a short note (3-5 lines) on your method and any data anomalies you noticed.

Bonus · Extract Insights from Our Python

OPTIONAL BUT VALUED

We've shared **winner.py** - the script we actually use to generate fRL winners. Read it (run or adapt it if you like), then tell us:

- In plain English, what the script does and which rules it encodes.
- How the New-vs-Existing split and the MVP score are calculated in the code.
- 2-3 genuine insights you'd draw from the winners or the underlying data.

Submit: a short write-up (half a page is plenty). Bullet points are fine.

Before you send

- [ ] Task 1 : agent link/screenshots + system prompt + 3 sample Q&A.
- [ ] Task 2 : fRL winners Excel + short method & anomalies note.
- [ ] Bonus : Python insights write-up.
- Email everything to **product.marketing@foundit.ai** Good luck!